Finanças II – Corporativas (MPE Insper)

Exercise list R3 — A frictionless benchmark, taxes, distress and static trade-off

Worked solutions

Statements are in `Listas/lista-r3-mm-taxes-tradeoff.md` and are not restated; each section cites the exercise by number and printed page. Editions cited are the local copies: Berk, DeMarzo & Harford, *Fundamentals of Corporate Finance*, 2nd ed. (2011); Ross, Westerfield, Jordan & Lamb, *Fundamentos de Administração Financeira*, 9th ed. (2013); Koski, Higgins & Mitton, *Analysis for Financial Management*, 13th ed. (2023); Grinblatt & Titman, *Financial Markets and Corporate Strategy*, 2nd ed. (2002); Amaro de Matos, *Theoretical Foundations of Corporate Finance* (2001). Every number below is reproduced by `lista-r3-mm-taxes-tradeoff-solutions_check.py`. Amounts are in dollars unless a section says "millions".

1. Two-outcome project financed with equity or safe debt — Berk 16.1, p. 490

Notation.

- $C_{\text{low}} = 130{,}000$, $C_{\text{high}} = 180{,}000$: project free cash flow at $t = 1$ in the two states; $p = \frac{1}{2}$: probability of the low state.
- $I = 100{,}000$: initial investment; $r_U = 0.20$: project (unlevered) cost of capital; $r_f = 0.10$: risk-free rate.
- V_U: value of the project's cash flows (= unlevered equity); $NPV = V_U - I$.
- $D = 100{,}000$: amount borrowed at r_f; E_L: value of levered equity; r_E: its expected return; $r_D = r_f$.

Step 1 — Expected cash flow.

$$\begin{aligned}\mathbb{E}[C] &= p\,C_{\text{low}} + (1-p)\,C_{\text{high}} \\ &= (0.5)(130{,}000) + (0.5)(180{,}000) \\ &= 155{,}000.\end{aligned} \tag{1.1}$$

Step 2 — (a) NPV. Discounting (1.1) at the project's cost of capital,

$$\begin{aligned}V_U &= \frac{\mathbb{E}[C]}{1+r_U} \\ &= \frac{(155{,}000)}{(1.20)} \\ &= 129{,}166.67,\end{aligned} \tag{1.2}$$

$$\begin{aligned}NPV &= V_U - I \\ &= (129{,}166.67) - (100{,}000) \\ &= 29{,}166.67.\end{aligned} \tag{1.3}$$

Step 3 — (b) Unlevered equity. All-equity holders receive exactly C_s, so by the law of one price their claim is worth (1.2): $E_U = V_U = 129{,}166.67$. Selling the equity raises \$129,166.67, more than I; the surplus is the NPV.

Step 4 — (c) Levered equity cash flows. The safe loan requires a repayment of

$$\begin{aligned}D(1+r_f) &= (100{,}000)(1.10) \\ &= 110{,}000,\end{aligned} \tag{1.4}$$

which is below the lowest C_s, so the debt is indeed riskless. Equity receives $C - 110{,}000$, i.e. $130{,}000 - 110{,}000 = 20{,}000$ or $180{,}000 - 110{,}000 = 70{,}000$.

Step 5 — (c) Value by MM I. With no taxes or distress costs, MM I gives $E_L + D = V_U$, so

$$\begin{aligned} E_L &= V_U - D \\ &= (129{,}166.67) - (100{,}000) \\ &= 29{,}166.67. \end{aligned} \tag{1.5}$$

Step 6 — Consistency with MM II. Expected equity cash flow is $(0.5)(20{,}000)+(0.5)(70{,}000) = 45{,}000$, so

$$\begin{aligned} r_E &= \frac{45{,}000}{E_L} - 1 \\ &= \frac{45{,}000}{(29{,}166.67)} - 1 \\ &= 0.542857, \end{aligned} \tag{1.6}$$

and MM II, $r_E = r_U + \frac{D}{E_L}(r_U - r_D)$, with $\frac{D}{E_L} = \frac{100{,}000}{29{,}166.67} = 3.428571$, gives

$$\begin{aligned} r_E &= (0.20) + (3.428571)(0.20 - 0.10) \\ &= 0.542857, \end{aligned}$$

the same as (1.6): levered equity is worth \$29,166.67 and earns 54.29%.

Intuition. Borrowing slices the same cash flows differently; it does not change their total value. The entrepreneur captures the same NPV, \$29,166.67, either as the excess of the equity sale over I or as the value of the levered equity left after the loan. The riskier levered equity simply demands a higher expected return.

Ref: Berk Ch. 16, §16.2 "Capital Structure in Perfect Capital Markets", pp. 464–472 (MM I, MM II).

2. One-time dividend or repurchase in a perfect market — Berk 17.7, p. 528

Notation.

- $N = 500$ million shares; $P_0 = 15$: cum-payout share price; $C = 250$ million: excess cash paid out.
- d: dividend per share; P_{ex}: ex-dividend price; n_R: shares repurchased (millions); P_R: post-repurchase price.

Step 1 — Market value of the firm. Equity (the firm has no debt) is worth $NP_0 = (500)(15) = 7{,}500$ million, cash included.

Step 2 — (a) Ex-dividend price.

$$d = \frac{C}{N} = \frac{(250)}{(500)} = 0.50, \tag{2.1}$$

$$\begin{aligned} P_{\text{ex}} &= \frac{NP_0 - C}{N} \\ &= \frac{(7{,}500) - (250)}{(500)} \\ &= 14.50. \end{aligned} \tag{2.2}$$

Step 3 — (b) Repurchase. In a perfect market the shares are bought at the fair price P_0:

$$n_R = \frac{C}{P_0} = \frac{(250)}{(15)} = 16.6667, \tag{2.3}$$

$$\begin{aligned} P_R &= \frac{NP_0 - C}{N - n_R} \\ &= \frac{(7{,}500) - (250)}{(500) - (16.6667)} \\ &= \frac{7{,}250}{483.3333} \\ &= 15.00. \end{aligned} \tag{2.4}$$

Step 4 — (c) Comparison. Per original share, the dividend leaves $P_{\text{ex}}+d = (14.50)+(0.50) = 15.00$ in stock plus cash, by (2.1)–(2.2); the repurchase leaves either a share worth 15.00 by (2.4) or 15.00 in cash for a tendered share. Wealth is \$15 per share under both policies: neither makes investors better off.

Intuition. Paying out cash shrinks the firm by exactly the cash paid. With a dividend the price falls by the dividend; with a repurchase the share count falls instead and the price stays put. Any investor can create the other pattern for herself (sell shares, or reinvest the dividend), so payout policy is irrelevant (MM).

Ref: Berk Ch. 17, §17.2 “Dividends Versus Share Repurchases in a Perfect Capital Market”, pp. 503–508.

3. MM I with corporate taxes — Ross 16.14, p. 578

Notation.

- $EBIT = 92{,}000$ per year forever; $R_U = 0.15$: unlevered cost of equity; $R_D = 0.09$: borrowing rate.
- $T_C = 0.35$: corporate tax rate; $D = 60{,}000$: perpetual debt used to buy back shares.
- V_U, V_L: unlevered and levered firm values; E: levered equity value.

Step 1 — Unlevered value. Cash flow to the all-equity firm is $EBIT(1-T_C) = (92{,}000)(1-0.35) = 59{,}800$ forever, so

$$\begin{aligned} V_U &= \frac{EBIT(1 - T_C)}{R_U} \\ &= \frac{(59{,}800)}{(0.15)} \\ &= 398{,}666.67. \end{aligned} \tag{3.1}$$

Step 2 — Value of the tax shield. The yearly saving is $T_C R_D D = (0.35)(0.09)(60{,}000) = 1{,}890$, a perpetuity as risky as the debt, hence discounted at R_D:

$$\begin{aligned} PV(\text{TS}) &= \frac{T_C R_D D}{R_D} \\ &= T_C D \\ &= \frac{(1{,}890)}{(0.09)} \\ &= 21{,}000. \end{aligned} \tag{3.2}$$

Step 3 — Levered value (MM I with taxes). Adding (3.2) to (3.1),

$$\begin{aligned} V_L &= V_U + T_C D \\ &= (398{,}666.67) + (21{,}000) \\ &= 419{,}666.67, \end{aligned} \tag{3.3}$$

of which equity is $E = V_L - D = (419{,}666.67) - (60{,}000) = 359{,}666.67$.

Intuition. The government's claim on EBIT falls by \$1,890 a year once interest is deductible; that perpetual saving, worth $T_C D$, accrues to the firm's investors. The borrowing rate does not affect the answer because it scales both the saving and its discount rate.

Ref: Ross Cap. 16, §16.4 "Proposições I e II de M&M com impostos sobre o lucro da pessoa jurídica", pp. 551–556.

4. APV with constant debt — Higgins 8.23, p. 333

Notation.

- $NPV_0 = 500{,}000$: base-case NPV of the project's free cash flows.
- $D = 1{,}000{,}000$: constant debt for $T = 3$ years; $r = 0.08$: interest and discount rate of the shields; $t = 0.40$: tax rate.
- TS: annual interest tax shield, received at the end of years 1–3; APV: adjusted present value.

Step 1 — Annual shield.

$$\begin{aligned} TS &= D\,r\,t \\ &= (1{,}000{,}000)(0.08)(0.40) \\ &= 32{,}000. \end{aligned} \tag{4.1}$$

Step 2 — Three-year annuity factor at 8%.

$$\begin{aligned} A &= \frac{1}{1.08} + \frac{1}{1.08^2} + \frac{1}{1.08^3} \\ &= (0.925926) + (0.857339) + (0.793832) \\ &= 2.577097. \end{aligned} \tag{4.2}$$

Step 3 — PV of the shields and APV. Using (4.1) and (4.2),

$$\begin{aligned} PV(\text{TS}) &= TS \cdot A \\ &= (32{,}000)(2.577097) \\ &= 82{,}467.10, \end{aligned} \tag{4.3}$$

$$\begin{aligned} APV &= NPV_0 + PV(\text{TS}) \\ &= (500{,}000) + (82{,}467.10) \\ &= 582{,}467.10. \end{aligned} \tag{4.4}$$

Book answer. Matches the book's answer, \$582,467 (p. 425).

Intuition. APV separates the value of the operations (\$500,000) from the value created by the financing side effect. With a fixed debt the shields are as safe as the debt, so they are discounted at the debt rate.

Ref: Higgins Ch. 8, Appendix "Asset Beta and Adjusted Present Value", pp. 318–325 (APV, p. 321).

5. MM II with a rising cost of debt — Berk 16.5, p. 491

Notation.

- $r_U = 0.12$: expected return on unlevered equity (= on the assets, unchanged by leverage without taxes).
- D/E: debt-equity ratio (0.50 or 1.50); r_D: debt cost of capital (0.06 or 0.08); r_E: expected return on levered equity.

Step 1 — MM II. With no taxes, the asset return is the value-weighted average of the claims, $r_U = \frac{E}{D+E} r_E + \frac{D}{D+E} r_D$; solving for r_E,

$$r_E = r_U + \frac{D}{E}(r_U - r_D). \tag{5.1}$$

Step 2 — (a) $D/E = 0.50$, $r_D = 6\%$. Substituting into (5.1),

$$\begin{aligned} r_E &= (0.12) + (0.50)(0.12 - 0.06) \\ &= 0.15. \end{aligned} \tag{5.2}$$

Step 3 — (b) $D/E = 1.50$, $r_D = 8\%$. Substituting into (5.1),

$$\begin{aligned} r_E &= (0.12) + (1.50)(0.12 - 0.08) \\ &= 0.18. \end{aligned} \tag{5.3}$$

Step 4 — Check: the WACC is unchanged. With $E/V = 1/1.5$ and $D/V = 0.5/1.5$ in (a), and $E/V = 1/2.5$, $D/V = 1.5/2.5$ in (b),

$$\begin{aligned} &\text{(a)} \quad \tfrac{1}{1.5}(0.15) + \tfrac{0.5}{1.5}(0.06) = 0.12, \\ &\text{(b)} \quad \tfrac{1}{2.5}(0.18) + \tfrac{1.5}{2.5}(0.08) = 0.12. \end{aligned}$$

Step 5 — (c) Reply to the manager. The higher expected return in (5.3) is pure compensation for higher risk: equity bears the same business risk on a thinner base. By MM I the firm's value, and therefore the share price, is the same at every debt level; the WACC stays at 12%. Choosing leverage to maximise r_E does not raise shareholder wealth; shareholders can also lever up on their own (homemade leverage) if they want a riskier, higher-return position. Shareholders care about the value of their claim, not its expected return.

Intuition. The cost of equity rises linearly with D/E at slope $r_U - r_D$. When debt becomes riskier (r_D rises), part of the risk moves to creditors and r_E rises less steeply, but the total required return on the assets is still r_U.

Ref: Berk Ch. 16, §16.2, pp. 464–472 (MM II, leverage and the cost of equity).

6. Tax shields on amortising debt — Berk 16.11, p. 492

Notation.

- B_t: debt outstanding during year $t = 1, \ldots, 4$ (millions); $B_1 = 100$, $B_{t+1} = B_t - 25$.
- $r_D = 0.10$: interest rate and discount rate of the shields; $\tau_c = 0.40$: marginal corporate tax rate.
- $TS_t = \tau_c r_D B_t$: tax shield received at the end of year t.

Assumption. The first \$25 million repayment is made one year from now, together with the first interest payment, and interest in year t is charged on the balance at the start of the year. The loan is then repaid after four years.

Step 1 — Schedule. Balances are $100, 75, 50, 25$, so interest $r_D B_t$ is $10, 7.5, 5, 2.5$ and

$$TS_t = (0.40)(0.10)B_t \in \{4,\ 3,\ 2,\ 1\}. \tag{6.1}$$

Step 2 — Present value. Discounting (6.1) at r_D (same risk as the loan),

$$\begin{aligned} PV(\text{TS}) &= \sum_{t=1}^{4} \frac{TS_t}{(1+r_D)^t} \\ &= \frac{(4)}{1.10} + \frac{(3)}{1.10^2} + \frac{(2)}{1.10^3} + \frac{(1)}{1.10^4} \\ &= 3.636364 + 2.479339 + 1.502630 + 0.683013 \\ &= 8.301346. \end{aligned} \tag{6.2}$$

The interest tax shields are worth about \$8.30 million.

Intuition. Each year's shield is proportional to the debt still outstanding, so amortisation shrinks the shield stream. A permanent \$100 million loan would be worth $\tau_c D = \$40$ million in shields; repaying it over four years leaves only about a fifth of that.

Ref: Berk Ch. 16, §16.3 "Debt and Taxes", pp. 473–479 (value of the interest tax shield).

7. Homemade leverage between twin firms — Ross 16.18, p. 578

Notation.

- $X = 400{,}000$: perpetual operating income of each firm (random, identical for both); no taxes.
- Weber (unlevered): $E_W = V_W = 2{,}500{,}000$. Cavaleiro (levered): $E_C = 1{,}632{,}000$, $D_C = 1{,}200{,}000$, interest $I_C = 72{,}000$ per year, so $r_D = 72{,}000/1{,}200{,}000 = 0.06$.
- $\alpha = 0.05$: fraction bought; $r = 0.06$: personal borrowing rate; L: amount borrowed personally.

Assumption. "Os custos líquidos das estratégias sejam iguais" is read as: both strategies require the same own funds today, the \$81,600 that 5% of Cavaleiro costs.

Step 1 — Strategy C: 5% of Cavaleiro. Equity income is $X - I_C = 400{,}000 - 72{,}000 = 328{,}000$, so

$$\text{cost}_C = \alpha E_C = (0.05)(1{,}632{,}000) = 81{,}600, \tag{7.1}$$

$$\text{income}_C = \alpha(X - I_C) = (0.05)(328{,}000) = 16{,}400, \tag{7.2}$$

a return of $16{,}400/81{,}600 = 20.10\%$.

Step 2 — Strategy W: 5% of Weber with the same own funds. The stake costs $(0.05)(2{,}500{,}000) = 125{,}000$ and earns $(0.05)(400{,}000) = 20{,}000$. With own funds (7.1),

$$L = 125{,}000 - 81{,}600 = 43{,}400, \tag{7.3}$$

$$\begin{aligned}\text{income}_W &= 20{,}000 - rL = (20{,}000) - (0.06)(43{,}400)\\ &= (20{,}000) - (2{,}604) = 17{,}396,\end{aligned} \tag{7.4}$$

a return of $17{,}396/81{,}600 = 21.32\%$.

Step 3 — (a) The return is higher at no extra risk. Both incomes contain $0.05X$, so they move one-for-one with the same operating risk; (7.4) exceeds (7.2) by $17{,}396 - 16{,}400 = 996$ in every state. Yes: the investor raises her return by buying Weber and borrowing, and the position is no riskier (it even uses less leverage: \$43,400 of personal debt against her 5% share of Cavaleiro's \$1,200,000, i.e. \$60,000).

Step 4 — The underlying arbitrage. Replicating strategy C exactly needs $L = \alpha D_C = (0.05)(1{,}200{,}000) = 60{,}000$: income $20{,}000 - (0.06)(60{,}000) = 16{,}400$, identical to (7.2), for own funds $125{,}000 - 60{,}000 = 65{,}000$. The saving is

$$81{,}600 - 65{,}000 = 16{,}600 = \alpha(V_C - V_W) = (0.05)(2{,}832{,}000 - 2{,}500{,}000), \tag{7.5}$$

and investing it at 6% yields $(0.06)(16{,}600) = 996$ a year, the gap found in Step 3.

Step 5 — (b) What investors do and when it stops. Every Cavaleiro shareholder gains by selling Cavaleiro shares and buying Weber shares with personal borrowing. Selling pressure lowers E_C and buying pressure raises E_W until the arbitrage profit (7.5) is zero, i.e. until $V_C = V_W$ (MM I). If Weber's value stays at \$2.5 million, it stops when $E_C = 2{,}500{,}000 - 1{,}200{,}000 = 1{,}300{,}000$; Cavaleiro's cost of equity is then $328{,}000/1{,}300{,}000 = 25.23\%$, which is MM II: $0.16 + \frac{1.2}{1.3}(0.16 - 0.06) = 25.23\%$, with $r_U = 400{,}000/2{,}500{,}000 = 0.16$.

Intuition. Investors can borrow on the same terms as the firm, so corporate leverage adds nothing they cannot build themselves. A levered firm that trades at a premium is selling something investors can make more cheaply at home, and arbitrage removes the premium.

Ref: Ross Cap. 16, §16.2 "O efeito da alavancagem financeira" (homemade leverage), pp. 542–547; §16.3, pp. 547–551 (Propositions I and II).

8. Announcement effect of a debt-financed repurchase with distress costs — Berk 16.22, p. 494

Notation.

- $P_0 = 5.50$, $N = 10$ million shares, so $E_0 = (5.50)(10) = 55$ million.
- $D = 20$ million of permanent new debt; $\tau_c = 0.30$; P_1: price after the announcement; $PV(\text{FDC})$: PV of financial distress costs.

Assumption. The announcement is credible and all the value change is priced at once; shares are later bought back at the post-announcement price, so the repurchase itself transfers no value.

Step 1 — (a) Taxes only. Permanent debt adds $\tau_c D = (0.30)(20) = 6$ million (cf. (3.2)), all of it to the current shareholders:

$$\begin{aligned} P_1 &= \frac{E_0 + \tau_c D}{N} \\ &= \frac{(55) + (6)}{(10)} \\ &= 6.10. \end{aligned} \tag{8.1}$$

Step 2 — (b) Taxes and distress costs. By the trade-off identity $V_L = V_U + \tau_c D - PV(\text{FDC})$, the observed equity gain is

$$\begin{aligned} (5.75)(10) - (55) = 2.5 &= \tau_c D - PV(\text{FDC}), \\ PV(\text{FDC}) &= (6) - (2.5) \\ &= 3.5 \text{ million}. \end{aligned} \tag{8.2}$$

Intuition. The market pays the shareholders for the expected tax savings and charges them for expected distress costs. The smaller price jump (\$0.25 against \$0.60) shows that the market expects distress costs to eat more than half of the tax benefit of this debt.

Ref: Berk Ch. 16, §16.3 pp. 473–479 and §16.5 "Optimal Capital Structure: The Tradeoff Theory", pp. 480–482.

9. After-tax WACC with riskless perpetual debt, then a debt-for-equity exchange — Grinblatt & Titman 13.15–13.16, p. 494

Notation.

- $D = 50$, $E = 50$ (millions): perpetual riskless debt and equity; $A = D + E = 100$: market value of the assets.
- $\beta_E = 1.2$; $r_f = 0.08$; $\mathbb{E}[r_M] = 0.13$; CAPM holds; $r_D = r_f$ (riskless debt); $T_c = 0.40$.
- UA, TX: unlevered assets and debt tax shield, $A = UA + TX$; β_{UA}, r_{UA}: their beta and expected return.
- Primes (D', E', V', $WACC'$) denote values after the exchange offer.

Step 1 — 13.15: cost of equity (CAPM).

$$\begin{aligned} r_E &= r_f + \beta_E(\mathbb{E}[r_M] - r_f) \\ &= (0.08) + (1.2)(0.13 - 0.08) \\ &= 0.14. \end{aligned} \tag{9.1}$$

Step 2 — 13.15: WACC. G&T's after-tax WACC (13.8) with (9.1):

$$\begin{aligned} WACC &= \frac{E}{D+E}\, r_E + \frac{D}{D+E}\, r_D(1-T_c) \\ &= (0.5)(0.14) + (0.5)(0.08)(1-0.40) \\ &= 0.094. \end{aligned} \tag{9.2}$$

Step 3 — Unlevered cost of capital (Hamada). With static perpetual riskless debt, $TX = T_c D$ (13.2) and $UA = D + E - T_c D = 100 - (0.40)(50) = 80$ (13.3). Unlevering the equity beta with (13.7),

$$\begin{aligned} \beta_{UA} &= \frac{\beta_E}{1 + (1-T_c)D/E} \\ &= \frac{(1.2)}{1 + (1-0.40)(1)} \\ &= 0.75, \end{aligned} \tag{9.3}$$

$$\begin{aligned} r_{UA} &= r_f + \beta_{UA}(\mathbb{E}[r_M] - r_f) \\ &= (0.08) + (0.75)(0.05) \\ &= 0.1175. \end{aligned} \tag{9.4}$$

Check with the adjusted cost of capital formula (p. 482), $WACC = r_{UA}(1 - T_c \frac{D}{D+E})$: $(0.1175)(1 - (0.40)(0.5)) = 0.094$, equal to (9.2).

Step 4 — 13.16: values after retiring \$25 million of debt. The unlevered assets do not change; the shield shrinks to $T_c D' = (0.40)(25) = 10$. Hence

$$\begin{aligned} V' &= UA + T_c D' = (80) + (10) = 90, \\ E' &= V' - D' = (90) - (25) = 65, \end{aligned} \tag{9.5}$$

of which \$25 million is the new shares given to the old creditors and $50 - (0.40)(25) = 40$ is the old shareholders' stake: they bear the \$10 million loss of shield value.

Step 5 — 13.16: new WACC. Substituting $D'/V' = 25/90 = 0.277778$ into the adjusted cost of capital formula with (9.4),

$$\begin{aligned} WACC' &= r_{UA}\Big(1 - T_c \frac{D'}{V'}\Big) \\ &= (0.1175)\big(1 - (0.40)(0.277778)\big) \\ &= (0.1175)(0.888889) \\ &= 0.104444. \end{aligned} \tag{9.6}$$

Step 6 — Cross-check through the cost of equity. By (13.6), $\beta'_E = \beta_{UA}(1 + (1-T_c)\frac{D'}{E'}) = (0.75)\big(1 + (0.6)\frac{25}{65}\big) = (0.75)(1.230769) = 0.923077$, so $r'_E = (0.08) + (0.923077)(0.05) = 0.126154$ and

$$\begin{aligned} WACC' &= \tfrac{65}{90}(0.126154) + \tfrac{25}{90}(0.08)(0.6) \\ &= (0.091111) + (0.013333) = 0.104444, \end{aligned}$$

the same as (9.6). The WACC rises from 9.40% to 10.44%.

Assumption. The debt ratio in (9.6) is measured at post-exchange market values. Using $D'/V = 25/100$ (ignoring the lost shield) would give $(0.1175)(1 - 0.40 \cdot 0.25) = 10.575\%$; that mixes the old firm value with the new debt and is not consistent with the Hamada model.

Intuition. In the Hamada model the only thing debt adds is a riskless tax shield. Cutting the debt removes part of that cheap, government-funded slice of the financing, so the WACC moves back toward the unlevered cost of capital, 11.75%.

Ref: Grinblatt & Titman Ch. 13: Hamada model, eqs. (13.1)–(13.7), pp. 463–466; WACC (13.8), p. 476; adjusted cost of capital formula, pp. 481–482.

10. Taxes vs. bankruptcy costs in two states — Grinblatt & Titman 16.2, p. 591

Notation.

- $C_G = 210$, $C_B = 66$: pre-tax cash flows next period in the favourable and unfavourable state, each with probability $\frac{1}{2}$.
- $t = \frac{1}{3}$: tax rate of the all-equity firm; $s = 0.05$: tax saved per dollar of debt financing; $r_f = 0.10$; investors are risk neutral.
- F: face value of the zero-coupon debt; D: its market value today; $K = 20$: bankruptcy cost; V_U, E, V: unlevered value, equity, levered firm value.

Assumption. "Debt financing used" is the amount raised today, D; the saving sD is realised next period in both states (if the firm defaults it accrues to the creditors who take it over). The answer to (e) does not depend on this reading (Step 6).

Step 1 — (a) All-equity value. After-tax flows are $210(1 - \frac{1}{3}) = 140$ and $66(1 - \frac{1}{3}) = 44$, so

$$\begin{aligned} V_U &= \frac{\frac{1}{2}(140) + \frac{1}{2}(44)}{1.10} \\ &= \frac{920}{11} \\ &= 83.6364. \end{aligned} \tag{10.1}$$

Step 2 — (b) Debt with $F = 44$. Guess the debt is safe: $D = 44/1.10 = 40$, so the saving is $sD = (0.05)(40) = 2$. In the bad state the firm then has $44 + 2 = 46 \geq 44$: no default, confirming the guess. The firm receives $D = 40$.

Step 3 — (c) Equity. Equity gets $140 + 2 - 44 = 98$ (good) and $46 - 44 = 2$ (bad):

$$\begin{aligned} E &= \frac{\frac{1}{2}(98) + \frac{1}{2}(2)}{1.10} \\ &= \frac{500}{11} \\ &= 45.4545. \end{aligned} \tag{10.2}$$

Step 4 — (d) Firm value.

$$\begin{aligned} V &= D + E \\ &= (40) + (45.4545) \\ &= 85.4545, \end{aligned} \tag{10.3}$$

equal to $V_U + \frac{2}{1.10} = \frac{920}{11} + \frac{20}{11} = \frac{940}{11}$: the PV of the tax saving, 1.8182, is the whole gain.

Step 5 — (e) Debt with $F = 70$. Now $44 + sD < 70$ in the bad state, so the firm defaults; creditors take over and receive $44 + 0.05D - 20 = 24 + 0.05D$. In the good state they get 70. Hence

$$\begin{aligned} D &= \frac{\frac{1}{2}(70) + \frac{1}{2}(24 + 0.05D)}{1.10} \\ 1.10\,D &= 47 + 0.025\,D \\ D &= \frac{47}{1.075} = 43.72093. \end{aligned} \tag{10.4}$$

With $sD = 2.18605$, the bad-state recovery is $26.18605 < 70$ (default confirmed) and equity gets $70 + 2.18605 = 72.18605$ in the good state and 0 in the bad state:

$$\begin{aligned} E &= \frac{\frac{1}{2}(72.18605)}{1.10} \\ &= 32.81184, \\ V &= D + E \\ &= (43.72093) + (32.81184) \\ &= 76.53277. \end{aligned} \tag{10.5}$$

Equivalently $V = V_U + PV(sD) - PV(K) = (83.63636) + (1.98732) - (9.09091) = 76.53277$, using $2.18605/1.10 = 1.98732$ and $\frac{1}{2}(20)/1.10 = 9.09091$. No: the firm is worth less with a \$70 obligation (76.53 against 85.45).

Step 6 — Robustness. Whatever the basis of the saving, it is at most 0.05×70 next period, worth 3.18182, while expected bankruptcy costs are worth 9.09091; so $V \leq (83.63636) + (3.18182) - (9.09091) = 77.72727 < 85.4545$ under any reading.

Intuition. This is the static trade-off in miniature. Safe debt adds a small tax subsidy at no cost. Debt large enough to default in the bad state buys a little more subsidy but triggers a deadweight loss of \$20 with probability one half, which dominates.

Ref: Grinblatt & Titman Ch. 16, §16.1 "Bankruptcy" (direct bankruptcy costs vs. the debt tax advantage), pp. 559–561.

11. Risky debt in a binomial market and new debt of equal seniority — Grinblatt & Titman 14.2–14.3, pp. 526–527

Notation.

- $C_G = 210$, $C_B = 80$: firm cash flow in the good/bad state; risk-neutral probability $\frac{1}{2}$ each; $r_f = 0$; no taxes.
- $F_0 = 50$: face of the existing debt; $F_1 = 50$: face of the new equal-seniority debt; 100 shares initially.
- V: firm (machine) value; D_0, D_1: market values of old and new debt; P: share price; n_R: shares repurchased.

Step 1 — 14.2: share price. With $r_f = 0$, values are risk-neutral expectations:

$$V = \tfrac{1}{2}(210) + \tfrac{1}{2}(80) = 145, \tag{11.1}$$

the old debt is safe ($80 \geq 50$), $D_0 = 50$, equity is $145 - 50 = 95$, and $P = 95/100 = 0.95$.

Step 2 — (a) Price of the new debt. Total face becomes $F_0 + F_1 = 100 > 80$, so in the bad state the \$80 is shared pro rata, 40 to each issue. The new debt is worth

$$D_1 = \tfrac{1}{2}(50) + \tfrac{1}{2}(40) = 45. \tag{11.2}$$

The old debt now has the same payoff, so D_0 also falls from 50 to 45.

Step 3 — (b) Share price at the announcement. The firm holds the machine plus the \$45 cash raised, and owes debt worth $2 \times 45 = 90$:

$$E = (145) + (45) - (90) = 100, \qquad P = \frac{100}{100} = 1.00. \tag{11.3}$$

The firm then buys back $n_R = 45/1.00 = 45$ shares; 55 remain, and equity is $145 - 90 = 55$, paying 110 or 0, worth $\frac{1}{2}(110) = 55$, i.e. still \$1.00 per share.

Assumption. "A shareholder with 20 percent of the shares" holds 20 of the 100 shares before the announcement; "undoing the leverage change" means rebuilding the pre-announcement payoff exposure with personal trades.

Step 4 — (c) Undoing the leverage. Before: 20% of an equity paying $(160, 30)$, i.e. $(32, 6)$, worth $\frac{1}{2}(32) + \frac{1}{2}(6) = 19$. After the announcement the 20 shares are worth 20. Keep $0.2 \times 55 = 11$ shares (paying $(22, 0)$) and sell $20 - 11 = 9$ shares at \$1.00; with the \$9 buy 20% of the new debt, $0.2 \times 45 = 9$, which pays $0.2 \times (50, 40) = (10, 8)$. The position pays

$$(22 + 10,\ 0 + 8) = (32,\ 8), \qquad \text{worth } \tfrac{1}{2}(32) + \tfrac{1}{2}(8) = 20. \tag{11.4}$$

This beats the old payoff $(32, 6)$ state by state; the gain $20 - 19 = 1$ is 20% of the \$5 lost by the old creditors. Alternatively, she can rebuild $(32, 6)$ exactly with 12.25 shares and 7.5% of all the debt (paying $0.075 \times (100, 80) = (7.5, 6)$), at a cost of $12.25(1.00) + 0.075(90) = 19.00$, and keep \$1 in cash.

Step 5 — (d) MM still holds. Total firm value is $50 + 95 = 145$ before and $90 + 55 = 145$ after, as (11.1) requires: changing the capital structure did not change the value of the claims taken together, and investors can undo it (Step 4). The share price rose only because equal-seniority debt diluted the old creditors, a transfer of \$5 from them to the shareholders, not value creation.

Intuition. MM is a statement about the size of the pie, not about how it is cut. Issuing new debt that ranks equally with old debt cuts the old creditors' slice; without protective covenants, shareholders pocket that slice.

Ref: Grinblatt & Titman Ch. 14, §14.1 "The Modigliani-Miller Theorem" p. 501, §14.2 "How an Individual Investor Can 'Undo' a Firm's Capital Structure Choice" p. 505, §14.3 "How Risky Debt Affects the Modigliani-Miller Theorem" pp. 506–508.

12. Leveraged recapitalisation with corporate taxes — Berk 16.18, p. 493

Notation.

- $N_0 = 20$ million shares at $P_0 = 7.50$; all equity before the announcement.
- $D = 50$ million of permanent interest-only debt; $\tau_c = 0.40$; P_1: share price after the debt is issued; n_R: shares repurchased (millions).

Step 1 — (a) Existing assets. With no debt, assets equal equity: $N_0 P_0 = (20)(7.50) = 150$ million.

Step 2 — (b) Assets after the issue, before the repurchase. The firm holds the old assets, the \$50 million cash, and the tax shield of permanent debt, $\tau_c D = (0.40)(50) = 20$:

$$A_1 = (150) + (50) + (20) = 220 \text{ million}. \tag{12.1}$$

Step 3 — (c) Price and shares repurchased. Equity is $A_1 - D = (220) - (50) = 170$ million, so

$$P_1 = \frac{170}{20} = 8.50, \tag{12.2}$$

$$n_R = \frac{D}{P_1} = \frac{(50)}{(8.50)} = 5.882353 \text{ million shares}. \tag{12.3}$$

Step 4 — (d) After the repurchase. The cash is gone:

Assets	(\$m)	Liabilities and equity	(\$m)
Existing assets	150	Debt	50
Tax shield	20	Equity	120
Total	170	Total	170

With $20 - 5.882353 = 14.117647$ million shares left, the price is $120/14.117647 = 8.50$, unchanged by the repurchase itself.

Intuition. The whole tax benefit, \$20 million or \$1 per share, is capitalised into the price at the announcement. Shareholders who sell in the buyback and those who stay share it equally, since the repurchase is at the fair price \$8.50.

Ref: Berk Ch. 16, §16.3 "Debt and Taxes", pp. 473–479 (recapitalising to capture the tax shield).

13. Derive the MM adjusted cost of capital — Ross 16.19, p. 578

Notation.

- $EBIT$: constant perpetual operating income, fully paid out; $T_C \in [0, 1)$: corporate tax rate (no personal taxes).
- D: perpetual riskless debt with rate R_D; E: levered equity value; $V = E + D = V_L$: levered firm value; V_U: unlevered value.
- R_U: unlevered cost of capital; R_E: cost of levered equity; $\text{CMPC} = \frac{E}{V}R_E + \frac{D}{V}R_D(1 - T_C)$ (Ross's WACC). Assume $E > 0$, $V > 0$.

Claim. $\text{CMPC} = R_U[1 - T_C(D/V)]$.

Step 1 — Valuation identities (Ross §16.4). The unlevered firm's perpetual cash flow is $EBIT(1 - T_C)$, so by definition of R_U

$$EBIT(1 - T_C) = R_U V_U. \tag{13.1}$$

The shield $T_C R_D D$ is a perpetuity as risky as the debt, worth $T_C D$ (cf. (3.2)), so MM I with taxes is

$$V = V_U + T_C D. \tag{13.2}$$

Step 2 — Returns of the claims. Shareholders receive $(EBIT - R_D D)(1 - T_C)$ forever and debtholders $R_D D$ (a perpetuity worth D at rate R_D); valuing the equity stream as a perpetuity gives

$$E\,R_E = (EBIT - R_D D)(1 - T_C). \tag{13.3}$$

Step 3 — Multiply the CMPC by V. From the definition,

$$\begin{aligned} V \cdot \text{CMPC} &= E\,R_E + D\,R_D(1 - T_C) \\ &= \big[(EBIT - R_D D)(1 - T_C)\big] + R_D D(1 - T_C) && \text{(substituting (13.3))} \\ &= EBIT(1 - T_C) \\ &= \big[R_U V_U\big] && \text{(substituting (13.1))} \\ &= R_U[V - T_C D] && \text{(substituting (13.2)).} \end{aligned} \tag{13.4}$$

Step 4 — Divide by $V > 0$.

$$\text{CMPC} = R_U\Big[1 - T_C \frac{D}{V}\Big]. \qquad \blacksquare \tag{13.5}$$

Step 5 — Same result through MM II. Substituting $EBIT(1 - T_C) = R_U(E + D - T_C D)$ (from (13.1) and (13.2)) into (13.3) and dividing by E gives MM II with taxes, $R_E = R_U + (R_U - R_D)\frac{D}{E}(1 - T_C)$. Then

$$\begin{aligned} \text{CMPC} &= \tfrac{E}{V}\big[R_U + (R_U - R_D)\tfrac{D}{E}(1 - T_C)\big] + \tfrac{D}{V}R_D(1 - T_C) \\ &= \tfrac{E}{V}R_U + \tfrac{D}{V}R_U(1 - T_C) \\ &= R_U\Big[\tfrac{E+D}{V} - T_C\tfrac{D}{V}\Big] = R_U\Big[1 - T_C\tfrac{D}{V}\Big], \end{aligned}$$

since $E + D = V$.

Step 6 — Numerical check with Exercise 3. There $T_C D/V = 21{,}000/419{,}666.67 = 0.05003971$, so CMPC $= (0.15)(1 - 0.05003971) = 0.14249404$, and $(0.14249404)(419{,}666.67) = 59{,}800.00 = EBIT(1 - T_C)$, as (13.4) requires. Directly: $R_E = 56{,}290/359{,}666.67 = 0.156506$ and $\frac{359{,}666.67}{419{,}666.67}(0.156506) + \frac{60{,}000}{419{,}666.67}(0.09)(0.65) = 0.142494$.

Intuition. The levered firm's investors, taken together, receive the unlevered cash flow $EBIT(1-T_C)$ *plus* the tax saving; the CMPC nets the saving out of the debt cost, so it must discount the unlevered cash flow to the levered value. Since V exceeds V_U by $T_C D$, the rate falls below R_U by exactly $R_U T_C D/V$. The formula relies on fixed perpetual debt; under a constant leverage ratio the shield is riskier and the formula changes.

Ref: Ross Cap. 16, §16.4, pp. 551–556 (Proposition I with taxes; taxes, WACC and Proposition II).

14. Miller's gain from leverage with personal taxes — Amaro de Matos 2.2.7 #1, p. 58

Notation.

- X: random operating income per period of two otherwise identical firms, u (unlevered) and l (levered), forever.
- B: market value of l's perpetual riskless debt, which pays rB per period; $r > 0$ is the bond rate.
- $\tau_c, \tau_s, \tau_b \in [0, 1)$: corporate tax rate (interest deductible), personal tax rate on equity income, personal tax rate on interest income.
- $S_u = V_u$, S_l: equity values; $V_l = S_l + B$; $\alpha \in (0, 1]$: fraction held; L: personal loan; $G_L = V_l - V_u$: gain from leverage.

Assumption. As in Amaro's §2.2 and Appendix answer, individuals borrow and lend at the bond rate r, and personal interest is taxed (lending) or deductible (borrowing) at τ_b; there are no transaction costs and no arbitrage.

Step 1 — Payoff of the levered equity. The firm pays corporate tax on $X - rB$, and the holder pays τ_s on what she receives:

$$Y_l = \alpha(X - rB)(1 - \tau_c)(1 - \tau_s). \tag{14.1}$$

Step 2 — Payoff of the unlevered equity.

$$Y_u = \alpha X(1 - \tau_c)(1 - \tau_s). \tag{14.2}$$

Step 3 — Size of the personal loan. A loan L at r costs $rL(1-\tau_b)$ per period after tax. Choose L so this equals the debt-service part of (14.1):

$$\begin{aligned} rL(1 - \tau_b) &= \alpha rB(1 - \tau_c)(1 - \tau_s) \\ L &= \alpha B \, \frac{(1 - \tau_c)(1 - \tau_s)}{1 - \tau_b}. \end{aligned} \tag{14.3}$$

Step 4 — Replication. Buying α of u and borrowing L pays, by (14.2) and (14.3),

$$\begin{aligned} Y_u - rL(1 - \tau_b) &= \alpha X(1 - \tau_c)(1 - \tau_s) - \left[\alpha rB(1 - \tau_c)(1 - \tau_s)\right] \\ &= \alpha(X - rB)(1 - \tau_c)(1 - \tau_s) = Y_l, \end{aligned}$$

in every state and every period. Its cost today is $\alpha S_u - L$.

Step 5 — Law of one price. If $\alpha S_l > \alpha S_u - L$, a holder of l's shares sells them, builds the replica and keeps the difference with unchanged future income. If $\alpha S_l < \alpha S_u - L$, a holder of u's

shares sells them, buys α of l's shares and lends L (after-tax interest $rL(1-\tau_b)$, restoring Y_u by Step 4 read backwards), again pocketing the difference. No arbitrage therefore requires

$$\begin{aligned} \alpha S_l &= \alpha S_u - L \\ S_l &= S_u - B\,\frac{(1-\tau_c)(1-\tau_s)}{1-\tau_b} \qquad \text{(substituting (14.3), dividing by } \alpha). \end{aligned} \tag{14.4}$$

Step 6 — Firm values and the gain. Adding B to both sides of (14.4) and using $V_l = S_l + B$, $V_u = S_u$:

$$\begin{aligned} V_l &= V_u + B - B\,\frac{(1-\tau_c)(1-\tau_s)}{1-\tau_b}, \\ G_L &= V_l - V_u = \Big[1 - \frac{(1-\tau_c)(1-\tau_s)}{1-\tau_b}\Big]B. \qquad \blacksquare \end{aligned} \tag{14.5}$$

The gain goes to the stockholders who decide to lever: they collect B at issue and keep shares worth S_l, a total of V_l instead of V_u.

Step 7 — Special cases. (i) $\tau_s = \tau_b$ (including no personal taxes): $G_L = \tau_c B$, the MM (1963) result, Amaro (2.7). (ii) Miller's equilibrium (2.11), $(1-\tau_c)(1-\tau_s) = 1-\tau_b$: $G_L = 0$, so no single firm gains from leverage. (iii) $(1-\tau_c)(1-\tau_s) > 1-\tau_b$: $G_L < 0$, and the firm should not borrow.

Book answer. The final expression matches the book's answer (p. 255). The intermediate line printed there, "$\alpha S_l = \alpha S_u + \alpha B\frac{(1-\tau_c)(1-\tau_s)}{(1-\tau_b)}$", has a sign typo: it must be a minus, as in (14.4), since the investor *borrows* that amount and the book's own next line, $V_l = V_u + B[1-\cdots]$, follows only with the minus.

Intuition. Debt moves a dollar of income from the corporate-plus-equity tax channel, where it keeps $(1-\tau_c)(1-\tau_s)$, to the interest channel, where it keeps $(1-\tau_b)$. Leverage creates value only to the extent that the second channel is less taxed than the first.

Ref: Amaro de Matos Ch. 2, §2.2 "Personal and Corporate Taxation", pp. 48–52 (Miller equilibrium, G_L on p. 52); Appendix C.2.2, p. 255.

15. MM I with perpetual debt, and when borrowing and lending rates differ — Amaro de Matos 2.1.5 #3–#4, p. 48

Notation.

- X: random operating income per period (the time average of the stream), identical for firms u (unlevered) and l (levered); any distribution.
- B: market value of l's perpetual riskless debt, paying rB per period; r: corporate interest rate.
- $\tau \in [0,1)$: corporate tax rate, interest deductible ($\tau = 0$ is the no-tax case); no personal taxes.
- $V_u = S_u$, $V_l = S_l + B$; $\alpha \in (0,1]$: fraction of a firm held by an investor.
- #4 only: $\bar{r}$ and $\underline{r}$: individual borrowing and lending rates, $\bar{r} > r > \underline{r} > 0$.

Assumption. In #3 individuals borrow and lend at r, as firms do; there are no transaction costs. In #4, as in the book's answer, $\tau = 0$ and individuals can lend only at $\underline{r}$ (they cannot buy l's bonds at yield r).

#3 — MM I with and without taxes

Step 1 — Payoffs. Shareholders of u and l receive, each period,

$$Y_u = X(1-\tau), \qquad Y_l = (X - rB)(1-\tau). \tag{15.1}$$

Step 2 — Replicating l's equity. Buy α of u and borrow $\alpha(1-\tau)B$ at r. By (15.1) the income is

$$\alpha X(1-\tau) - r\,\alpha(1-\tau)B = \alpha(X - rB)(1-\tau) = \alpha Y_l,$$

at a cost of $\alpha S_u - \alpha(1-\tau)B$ today.

Step 3 — Upper bound. If $S_l > S_u - (1-\tau)B$, a holder of α of l's shares sells them for αS_l, builds the replica of Step 2 and keeps, with identical future income,

$$\begin{aligned}\alpha\big[S_l - S_u + (1-\tau)B\big] &= \alpha\big[(V_l - B) - V_u + (1-\tau)B\big] \\ &= \alpha\big[V_l - V_u - \tau B\big] > 0,\end{aligned}$$

an arbitrage. Hence $V_l \le V_u + \tau B$.

Step 4 — Replicating u's equity. Buy α of l's shares and $\alpha(1-\tau)$ of its debt (or lend $\alpha(1-\tau)B$ at r):

$$\alpha(X - rB)(1-\tau) + \alpha(1-\tau)rB = \alpha X(1-\tau) = \alpha Y_u,$$

at a cost of $\alpha\big[S_l + (1-\tau)B\big]$.

Step 5 — Lower bound. If $S_u > S_l + (1-\tau)B$, i.e. $V_l < V_u + \tau B$, a holder of u's shares sells them, buys the replica of Step 4 and keeps $\alpha[V_u + \tau B - V_l] > 0$. Hence $V_l \ge V_u + \tau B$.

Step 6 — MM I. Steps 3 and 5 give

$$V_l = V_u + \tau B \quad \text{(with taxes)}, \qquad V_l = V_u \quad (\tau = 0). \qquad \blacksquare \tag{15.2}$$

Book answer. Matches the book's answer (pp. 252–253), $V_u + \tau B = V_l$. The book's switching portfolio invests all the proceeds in the other firm instead of replicating exactly, which leads to the same two inequalities.

#4 — Different borrowing and lending rates ($\tau = 0$)

Step 7 — Replicating l's equity with costly borrowing. The replica must still hold α of u (the only other claim on X), so its debt service has to equal $\alpha r B$ at the rate $\bar{r}$:

$$\bar{r}\,\bar{L} = \alpha r B \quad \Longrightarrow \quad \bar{L} = \alpha B\,\frac{r}{\bar{r}}, \tag{15.3}$$

which gives income $\alpha X - \alpha r B = \alpha Y_l$ at a cost of $\alpha\big(S_u - B\,r/\bar{r}\big)$. (Borrowing the full αB would cost $\alpha\bar{r}B > \alpha r B$ per period: not a replica.)

Step 8 — Upper bound. No arbitrage against an overvalued l requires $S_l \le S_u - B\,r/\bar{r}$; adding B,

$$V_l \le V_u + B\Big(1 - \frac{r}{\bar{r}}\Big) = V_u + B\,\frac{\bar{r} - r}{\bar{r}}. \tag{15.4}$$

Step 9 — Replicating u's equity with poor lending. Buy α of l's shares and lend $\underline{L}$ at $\underline{r}$ with

$$\underline{r}\,\underline{L} = \alpha r B \quad \Longrightarrow \quad \underline{L} = \alpha B\,\frac{r}{\underline{r}} > \alpha B, \tag{15.5}$$

giving income $\alpha(X - rB) + \alpha r B = \alpha Y_u$ at a cost of $\alpha\big(S_l + B\,r/\underline{r}\big)$.

Step 10 — Lower bound. No arbitrage against an overvalued u requires $S_u \le S_l + B\,r/\underline{r}$, i.e. $V_u \le V_l - B + B\,r/\underline{r}$:

$$V_l \ge V_u - B\,\frac{r - \underline{r}}{\underline{r}}. \tag{15.6}$$

Step 11 — What happens to MM1. Combining (15.4) and (15.6),

$$V_u - B\,\frac{r-\underline{r}}{\underline{r}} \;\le\; V_l \;\le\; V_u + B\,\frac{\bar{r}-r}{\bar{r}}. \qquad \blacksquare \tag{15.7}$$

MM1 no longer pins down V_l: arbitrage only confines it to a band around V_u whose width grows with the debt B and with the spreads. With $\bar{r} = r + \epsilon$, $\underline{r} = r - \epsilon$ the band is $V_u - \epsilon B/(r-\epsilon) \le V_l \le V_u + \epsilon B/(r+\epsilon)$, and it collapses to $V_l = V_u$ as $\epsilon \to 0$. Leverage can now raise firm value (up to the upper bound): the firm borrows more cheaply than individuals, so corporate leverage is a service investors cannot fully copy. If individuals could buy l's bonds at yield r, Step 4 would apply and the lower bound would tighten to $V_l \ge V_u$.

Book answer. Differs from the book's printed band (p. 254), $V_u(1 - \epsilon B_l/X) \le V_l \le V_u(1 + \epsilon B_l/X)$. The qualitative conclusion is the same (a band around V_u, proportional to ϵB, vanishing at $\epsilon = 0$), but the bounds differ. The book's portfolios keep the investor's capital fixed (all proceeds go into the other firm, and the full principal B_l is borrowed or lent), so the two incomes differ by a multiple of the random X minus a fixed amount. Its inequality therefore holds only for a given value of X (or in expectation), not state by state. The portfolios above match incomes exactly, so (15.7) is a true no-arbitrage band for any distribution of X. The two coincide only if X/V_u happens to equal $\bar{r}$ (upper bound) or $\underline{r}$ (lower bound).

Intuition. MM1 rests on investors being able to copy corporate leverage on the same terms. When personal borrowing is dearer and lending cheaper than corporate debt, homemade leverage is an imperfect substitute: arbitrage still limits mispricing, but only to within the cost of the personal-borrowing wedge.

Ref: Amaro de Matos Ch. 2, §2.1.1–2.1.3, pp. 40–46 (MM1, MM2, MM with taxes); Appendix C.2.1, pp. 252–254.